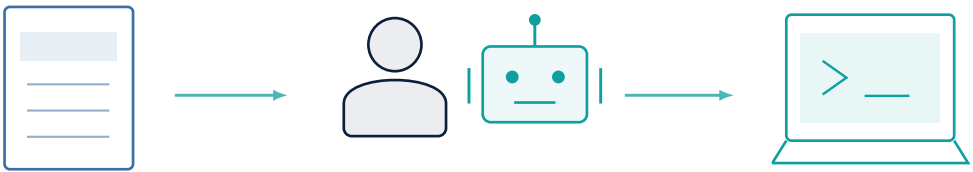
THE ESTIMATOR

Nguyen My Thong / AI-assisted build walkthrough



Frame the problem

Build with coding agents

Verify the result

OCT 16–19
Initial build + deployment

OCT 22
Fest Night

OCT 23–24
Later missing-data correction

01 / ASSUMPTIONS

Prototype scenario, not an observed Anduin workflow.



UNIT

Person-days ≠ delivery date

REPRESENTATION

15 features / sufficiency unvalidated

HISTORY

240 synthetic projects / no company data

LEARNING DEPENDENCY

Consistent scope + actual-effort records

02 / THE WORKING PRODUCT

Language model: interpret text. Calculator: estimate effort.

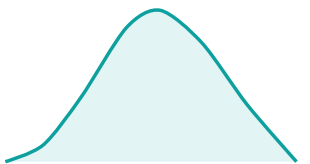
CLIENT EMAIL

"Mobile payments.
No multi-currency."

EXTRACT + QUOTES

- ✓ Mobile: true
- ✗ Multi-currency: false
- ? Migration: unknown

EFFORT RANGE



P10 / P50 / P90

Ask → answer → refine

Human review before commitment

03 / CONTRACTS BEFORE CODE

Specification and API contract existed in the initial commit.

EXTRACTION CONTRACT

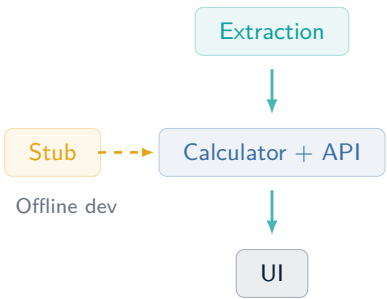
```
feature:  canonical_name
value:  true | false | null
evidence_quote:  source text
```

present

explicitly absent

not mentioned = unknown

ONE SHARED RESPONSE SHAPE



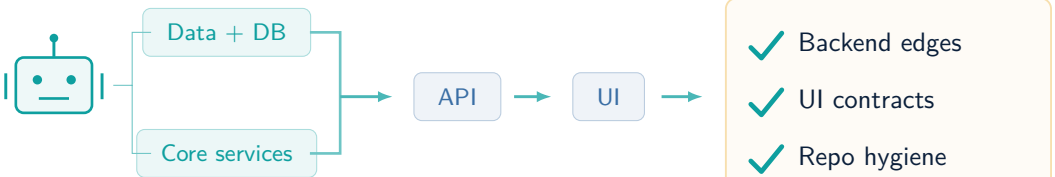
Gain: traceable, testable modules

Cost: schema + integration work

04 / AI-ASSISTED IMPLEMENTATION

OpenCode = coding agents. DeepSeek = runtime extraction.

BOUNDED MODULES | **INTEGRATION** | **QA + DEPLOY**



Separate files / frozen interfaces

ID collisions / invalid actuals

Cloud lacked .git

One service

05 / A REAL AGENT INTERACTION

TAWOS / a dataset mismatch became a narrower question.



ME / INITIAL REQUEST

Try evaluating the estimator on TAWOS.

AGENT / CONSTRAINT

Fields and labels do not fit.

Full evaluation is not supported.





ME / FOLLOW-UP

Which part can we still test?

Narrow the experiment, not the standard of evidence.

06 / TAWOS: LIMITED EXPERIMENT

Real Jira issues / story points, not person-days.



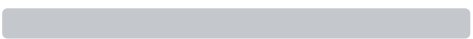
5,850 issues

39 projects

→ Offline stub → Refit + unseen projects

CROSS-PROJECT MAE / STORY POINTS



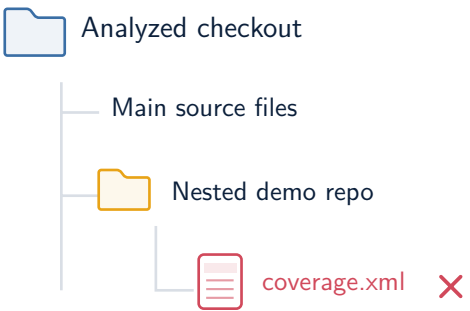


Vocabulary gap / 22% trigger any feature

Day-based floor ≠ story points

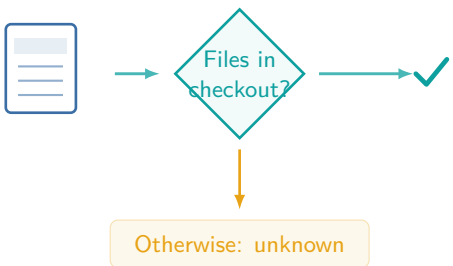
07 / INTEGRATION FAILURE: D28

Local and cloud scores differed / trace the evidence to its source.



Valid XML / evidence about the wrong repo

TARGETED FIX



Regression test + local/cloud match

08 / ASSUMPTION FAILURE: D31

Post-Fest, Oct 23–24 / external review challenged the missing-data assumption.

OVERALL FEATURE CELLS



25%

positive overall

EXTRACTION-NULL CELLS



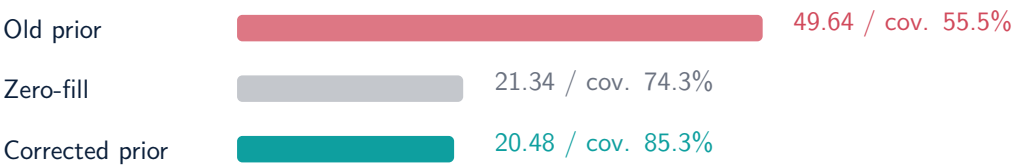
4.1%

positive among nulls

Randomly hidden ≠ not mentioned / use a null-conditioned prior

vs zero-fill: higher coverage / similar MAE / extra prior maintenance

MAE / PERSON-DAYS



09 / WHAT THE EVIDENCE SUPPORTS

Implemented behavior, controlled evaluation, and business value are different claims.



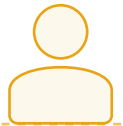
Working prototype

242 offline tests
Real-provider extraction eval



Synthetic replay

Extracted MAE: 20.48
De-biased team: 18.26
No human-superiority claim



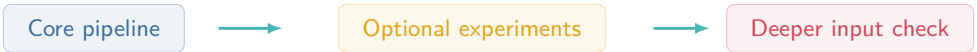
Real workflow pilot

Handling time / quality
Adoption / confidential-data policy
Still to establish

10 / WHAT I WOULD CHANGE

Keep the contracts / move evidence and user feedback earlier.

BUILD LESSON



More scope / more review surface

NEXT TIME



Simple baseline stays visible

Measure review + rework, not just generation

11 / APPLYING THE METHOD AT ANDUIN

Proposed approach / learn the workflow and agree priorities with the team.



Observe one workflow

Owner / cases / baseline



Build one slice

AI / script / SQL / rules



Pilot with review

Approved data / exceptions

Time + quality?

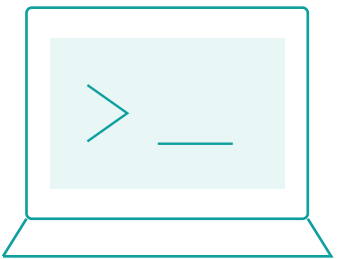
Keep + hand off

Simplify or stop

Document operation, fallback, ownership

12 / CLOSE

Nguyen My Thong / Questions



Frame

Build

Verify

github.com/iknizzz1807/the-estimator

APPENDIX / METHODS

Separate input uncertainty from effort uncertainty.



UNKNOWN INPUTS

2,000 seeded Monte Carlo draws

Feature-specific null priors
Independent sampling / simplifying assumption

PREDICTION ERROR

OOF residuals + tail widening

Not textbook split conformal
No formal coverage guarantee under drift

APPENDIX / METRICS

Synthetic effort history / person-days. TAWOS story-point scores are separate.

ALL 218 CHRONOLOGICAL PREDICTIONS	MAE	COVERAGE
Recorded features	17.36	86.7%
Extracted features / conditional prior	20.48	85.3%
De-biased simulated team	18.26	–

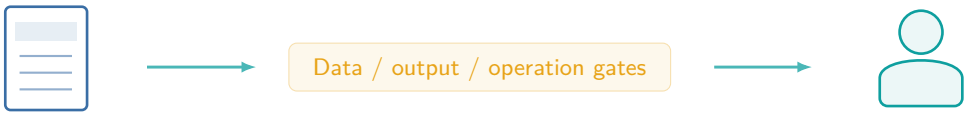
FINAL 42 STEPS / DIFFERENT INPUT INFORMATION

LLM-only 258.12 / LLM + history 35.27 / recorded-feature estimator 19.25

240 total / 228 completed. Extraction F1: 0.933 on 236/240 emails. Estimated batch cost: $0.20.

APPENDIX / RELEASE BOUNDARIES

Public demo ≠ approved confidential-document environment.



DATA + ACCESS

Auth + rate limits missing
Storage is ephemeral
Provider policy needs approval

OUTPUT + LEARNING

Unsupported values can remain
Add review + versioning
Review training labels

OPTIONAL CODEBASE LAB

Python-focused signals
Effort factors: demo-assumed
Preserve base result on failure

Null-prior rebuilding is separate from actual-effort retraining. Manual fallback and ownership need explicit agreement.